

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

88yGyDdvVRHcVsZF6obPaV3Qov7oX4L1HzAA15uboop

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Compounds: Xanax, Adderall

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Adanax

Formula: C17H12ClN5O

Weight: 412.86

Drug-likeness: 0.65

Activity:

CNS stimulant with anxiolytic properties

Target Analysis

Dopamine Transporter (DAT)

Binding Affinity: 8.2

Targeting DAT increases dopamine levels, beneficial for attention-deficit disorders.

GABA-A Receptor

Binding Affinity: 7.6

Enhancing GABAergic activity can reduce anxiety levels.

Toxicity Profile

Risk Level: Moderate

Affected Systems: Heart, Liver, CNS

Mitigation Suggestions:

- Regular monitoring of heart function
- Dose adjustment
- Avoid co-administration with other CNS depressants

Pharmacokinetics

Absorption:

Rapid oral absorption

Distribution:

Wide distribution, crosses the blood-brain barrier

Metabolism:

Hepatic metabolism via CYP3A4

Excretion:

Renal excretion

Half-life: 6-8 hours

Detailed Analysis

Adanax is a novel compound designed to merge the stimulatory effects of Adderall on the dopamine system with the anxiolytic effects of Xanax on the GABAergic system. The hypothesized mechanism includes inhibition of dopamine and norepinephrine reuptake, which enhances cognitive focus and attention, while simultaneously stimulating the GABA-A receptor, which can mediate anxiety reduction. Its chemical structure is derived from active moieties of both Adderall and Xanax, allowing it to cross the blood-brain barrier quickly and exhibit a long-lasting effect. Despite the favorable therapeutic effects, the dependency potential and increased risk of CNS depression indicate a need for cautious dosing, close monitoring, and a robust clinical trial design to ensure safety and efficacy.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

<https://ALCHEMUS.ai>